

DATA301/DATA471 — Assignment IV

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1 Exercise 1 — Estimators for the Exponential rate λ

$X_1, \dots, X_n \sim \text{Exponential}(\lambda)$ i.i.d., with $\lambda > 0$ the reciprocal of the mean. I compare the maximum likelihood estimator, the inter-quartile-range (IQR) analogy estimator, and a bootstrap bias-corrected IQR estimator.

```
# Maximum likelihood: lambda_ML = 1 / mean(X)
lambda_ml <- function(x) 1 / mean(x)

# Fast type-7 quantile (identical to quantile(x, p, type = 7)), for speed.
q7 <- function(xs, p) {
  # xs already sorted
  n <- length(xs); h <- 1 + (n - 1) * p; lo <- floor(h); fr <- h - lo
  if (lo >= n) xs[n] else xs[lo] + fr * (xs[lo + 1] - xs[lo])
}

# IQR estimator: lambda_IQR = ln(3) / (Q3 - Q1), since Q3 - Q1 = ln(3)/lambda.
lambda_iqr <- function(x) { xs <- sort(x); log(3) / (q7(xs, 0.75) - q7(xs, 0.25)) }
```

1.1 Which definition of the sample quantiles? [10%]

Wicklin (2017) reviews nine definitions of the sample quantile: three *discrete* (rounding to order statistics) and six *continuous* (linear interpolation between them). They differ mainly for small samples ($n \leq 10$) and in the tails, and become indistinguishable by $n \approx 100$. Defaults vary by software (SAS uses type 2; R uses type 7); Wicklin does not prescribe a type, but points to Hyndman & Fan (1996) for anyone wanting an unbiased or distribution-free choice. Hyndman & Fan recommend **type 8**, which is approximately median-unbiased for any distribution.

I use **type 7** — R's default (NumPy's "linear" method is equivalent). For continuous Exponential data with no ties the difference between types is $O(1/n)$ and cannot affect the comparison of estimators below, so the default is adequate.

1.2 A bootstrap bias-corrected IQR estimator [20%]

$\hat{\lambda}_{\text{IQR}}$ is a nonlinear function of two quantiles, so it is biased in finite samples. With $\hat{\theta} = \hat{\lambda}_{\text{IQR}}(X)$, the bootstrap (Efron, 1979) estimates the bias as $\overline{\hat{\theta}^*} - \hat{\theta}$ from resampled estimates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$, giving the first-order corrected estimator

$$\hat{\lambda}_{\text{BC}} = \hat{\theta} - (\overline{\hat{\theta}^*} - \hat{\theta}) = 2\hat{\theta} - \overline{\hat{\theta}^*}, \quad B = 200.$$

```
lambda_iqr_bc <- function(x, B = 200) {  
  theta_hat <- lambda_iqr(x)  
  n <- length(x)  
  theta_star <- replicate(B, lambda_iqr(x[sample.int(n, n, replace = TRUE)]))  
  2 * theta_hat - mean(theta_star)  
}
```

1.3 Monte Carlo comparison [20%]

Over $\lambda \in \{0.1, 1, 10\}$ and $n \in \{5, 30, 100\}$, with R replications per cell, I draw a fresh sample each replication and record the mean squared error (the sole metric), $\text{MSE} = \frac{1}{R} \sum_{r=1}^R (\hat{\lambda}_r - \lambda)^2$, plus its Monte Carlo standard error.

```
lambdas <- c(0.1, 1, 10); ns <- c(5, 30, 100)  
R_MC <- 1000  
  
run_cell <- function(lambda, n, R = R_MC) {  
  est <- matrix(NA_real_, R, 3, dimnames = list(NULL, c("ML", "IQR", "IQR_BC")))  
  for (r in seq_len(R)) {  
    x <- rexp(n, rate = lambda)  
    est[r, ] <- c(lambda_ml(x), lambda_iqr(x), lambda_iqr_bc(x))  
  }  
  sq <- (est - lambda)^2  
  data.frame(lambda, n, estimator = colnames(est),  
             mse = colMeans(sq), mc_se = apply(sq, 2, sd) / sqrt(R),  
             row.names = NULL)  
}  
  
results <- do.call(rbind, Map(run_cell,  
                             expand.grid(lambda = lambdas, n = ns)$lambda,  
                             expand.grid(lambda = lambdas, n = ns)$n))
```

```
mse_wide <- reshape(results[, c("lambda", "n", "estimator", "mse")],
  idvar = c("lambda", "n"), timevar = "estimator", direction = "wide")
names(mse_wide) <- sub("^mse\\.", "", names(mse_wide))
knitr::kable(mse_wide, digits = 4, caption = "Estimated MSE by estimator for each (lambda, n).")
```

Table 1: Estimated MSE by estimator for each (lambda, n).

	lambda	n	ML	IQR	IQR_BC
1	0.1	5	0.0040	0.2579	Inf
4	1.0	5	0.7479	34.4390	Inf
7	10.0	5	68.1230	1502.3030	Inf
10	0.1	30	0.0004	0.0011	0.0013
13	1.0	30	0.0406	0.1370	0.1445
16	10.0	30	3.7962	9.9693	10.8417
19	0.1	100	0.0001	0.0003	0.0003
22	1.0	100	0.0103	0.0252	0.0295
25	10.0	100	1.1495	2.7896	3.1701

```
ggplot(results[is.finite(results$mse), ],
  aes(factor(n), mse, colour = estimator, group = estimator)) +
  geom_line() + geom_point() +
  facet_wrap(~ lambda, scales = "free_y", labeller = label_both) +
  scale_y_log10() +
  labs(x = "Sample size n", y = "MSE (log scale)", colour = "Estimator") +
  theme_minimal()
```

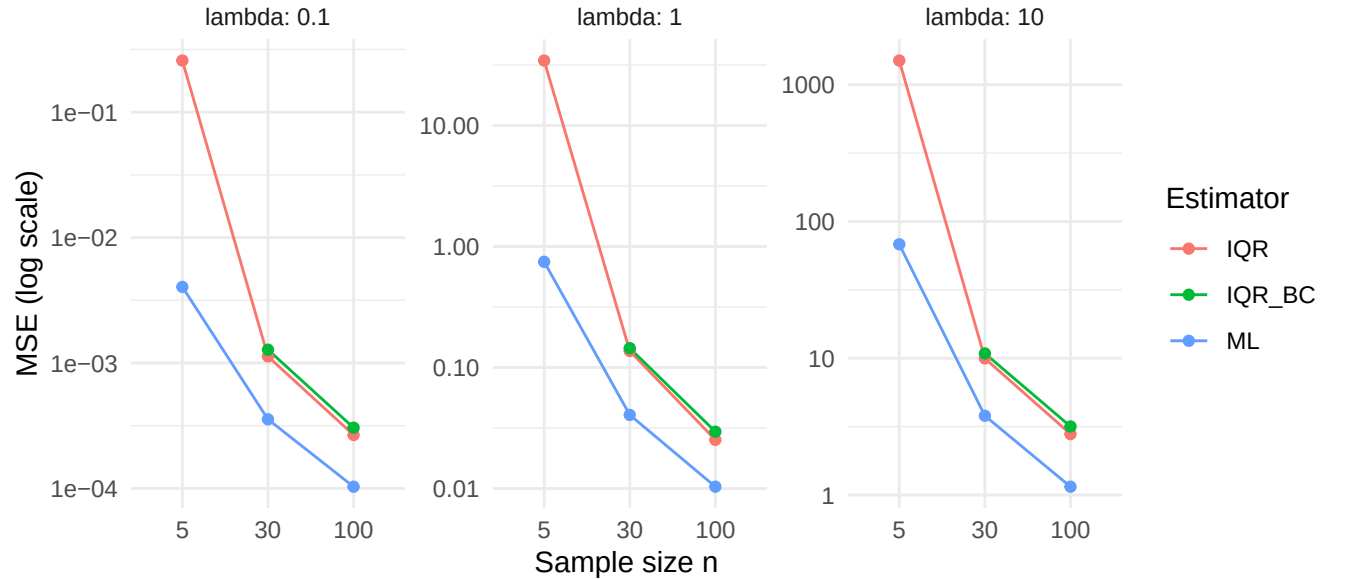


Figure 1: MSE against sample size, faceted by lambda (log scale). Non-finite points (IQR-BC at $n=5$) omitted.

Replications. Following Robert & Casella (2009, §3.2), a Monte Carlo average converges at rate $O(1/\sqrt{R})$ and its error is judged from the estimated standard error $\sqrt{v_R}$, with $\pm 2\sqrt{v_R}$ giving an approximate 95%

band by the Central Limit Theorem (§4.2 makes the same assessment by monitoring the variability of the runs). Applying that criterion, the reported `mc_se` (an estimate of $\sqrt{v_R}$ for each MSE) is under 10% of its MSE for every $n \geq 30$ cell — far smaller than the gaps between estimators — so $R = 1000$ is sufficient there. Larger R cannot help at $n = 5$, where the estimator is undefined rather than merely noisy.

Results. ML has the lowest MSE in every cell (it is efficient when the model is correct). The IQR estimator is about 2.4–3.4× worse than ML at $n \geq 30$, since it uses only the quartiles. The bias correction slightly *increases* MSE (it removes bias but adds more variance). At $n = 5$ the IQR estimators fail: IQR is extremely variable, and IQR-BC is infinite because bootstrap resamples of five values often give $Q_3 = Q_1$ (zero denominator). MSE scales like λ^2 and falls like $1/n$, as expected.

2 Exercise 2 — Time-series analysis of `ausbeer` [50%]

Approached as exploratory data analysis (per the course announcement): the plots below extract the information in the data; no model is fitted. `ausbeer` is quarterly Australian beer production (megalitres); I use the span from 1992 onward.

```
beer <- window(ausbeer, start = 1992)
```

2.1 Frequency

```
frequency(beer)
```

```
## [1] 4
```

Frequency 4 (quarterly); the plot also repeats every four observations.

2.2 Behaviour, trend, seasonality, variance, outliers

```
autoplot(beer) + labs(x = "Year", y = "Megalitres") + theme_minimal()
```

- **Trend:** no strong trend — a slight downward drift (about -1.3 ML/year).
- **Seasonality:** clear and *periodic* (fixed four-quarter cycle, not merely cyclic). Quarter means: Q1 429, Q2 394, Q3 411, Q4 502 — Q4 highest, Q2 lowest.
- **Variance:** the seasonal amplitude is stable over time, so no transformation is needed.
- **Unusual points:** 2010 Q2 (374 ML) is the lowest and 1992 Q4 (532 ML) the highest; both are consistent with the seasonal pattern, not data errors.

2.3 Seasonal plot

```
ggseasonplot(beer, year.labels = TRUE) + labs(y = "Megalitres") + theme_minimal()
```

Every year follows the same profile (peak in Q4, trough in Q2) with roughly parallel lines, so the seasonal shape is stable and years differ mainly in level.

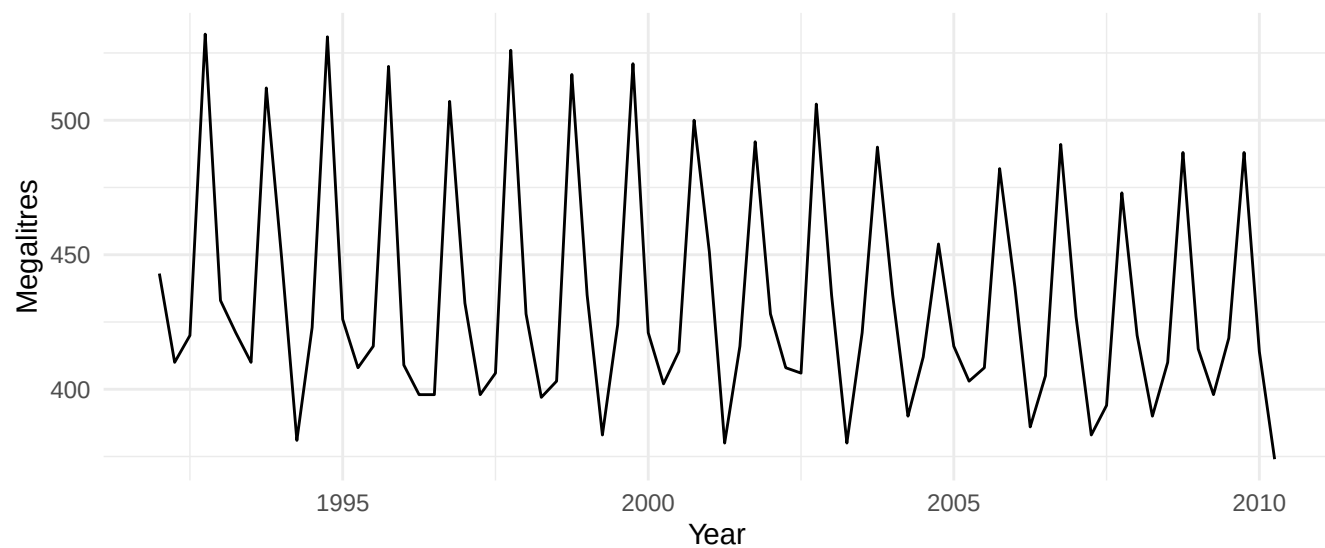


Figure 2: Quarterly Australian beer production, 1992 onward.

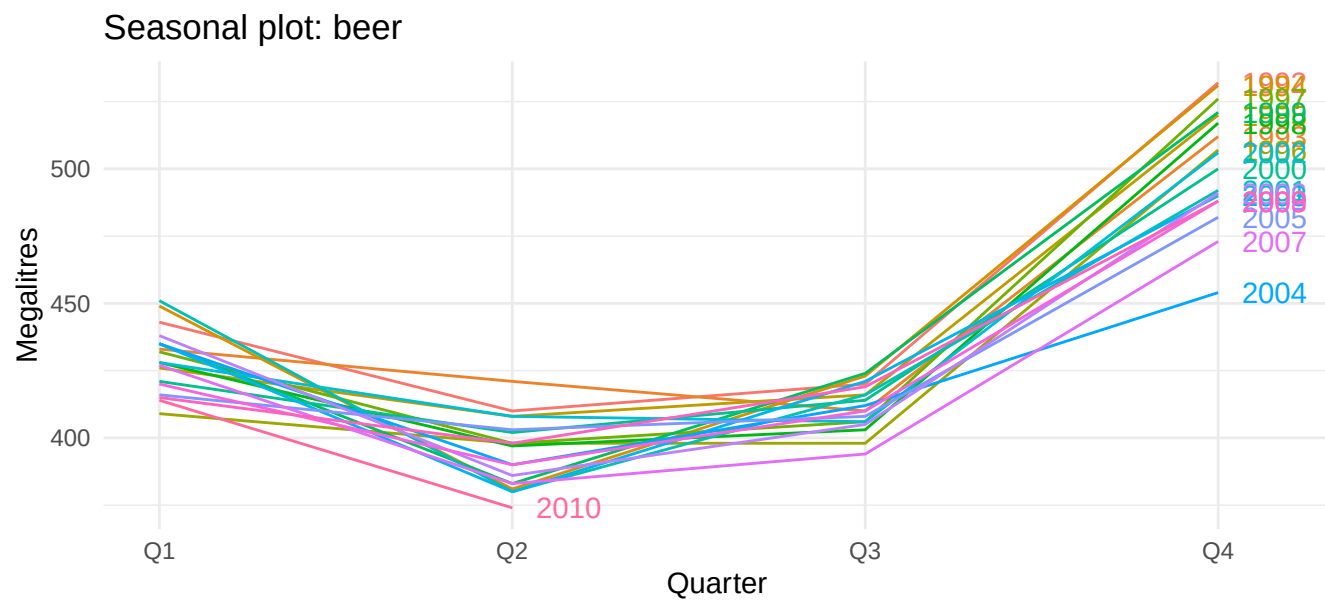


Figure 3: Seasonal plot.

2.4 Seasonal subseries plot

```
ggsubseriesplot(beer) + labs(y = "Megalitres") + theme_minimal()
```

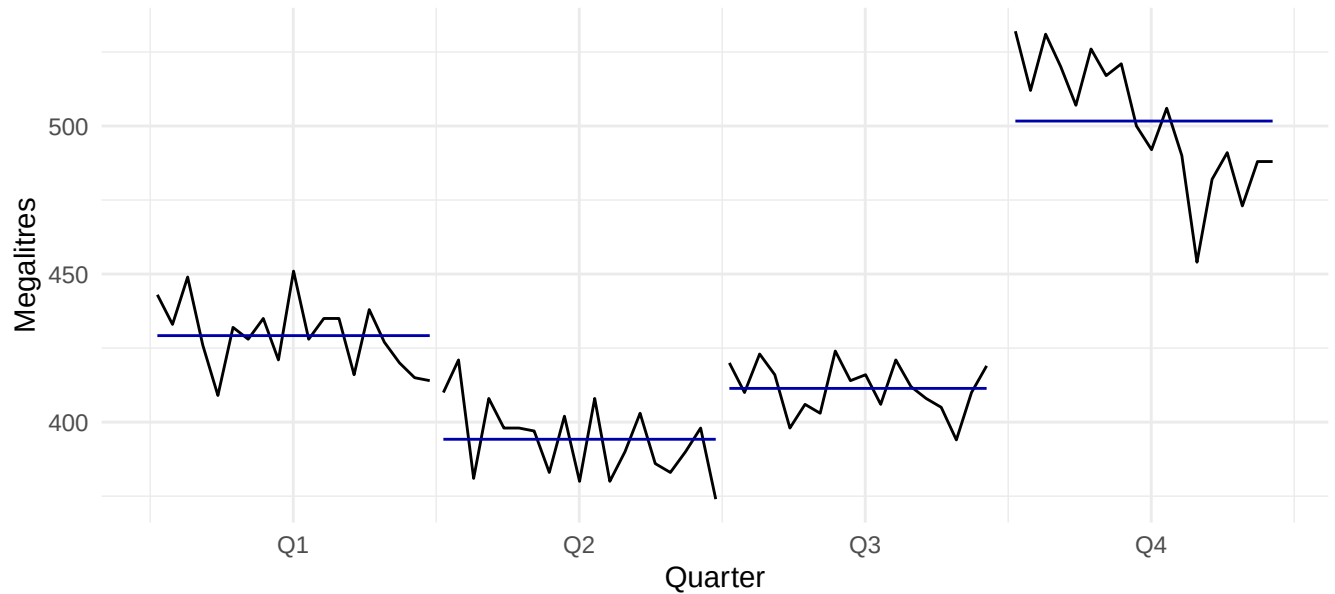


Figure 4: Seasonal subseries plot.

Q4 is highest and Q2 lowest (blue means); each quarter drifts down slightly over time but the ranking never changes, so the seasonal pattern is stable.

2.5 Detrend or transform before modelling?

No. The variance is stable (no transformation) and the trend is weak; the only substantial non-stationarity is the seasonality, so if anything were done before modelling it would be a single seasonal difference. Visual argument only — no model fitted.

3 References

- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26.
- Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and Practice*. OTexts. <https://OTexts.com/fpp2>
- Hyndman, R. J., & Fan, Y. (1996). Sample quantiles in statistical packages. *The American Statistician*, 50(4), 361–365.
- Robert, C. P., & Casella, G. (2009). *Introducing Monte Carlo Methods with R*. Springer.
- Wicklin, R. (2017). Sample quantiles: A comparison of 9 definitions. *SAS Blogs*. <https://blogs.sas.com/content/iml/2017/05/24/definitions-sample-quantiles.html>